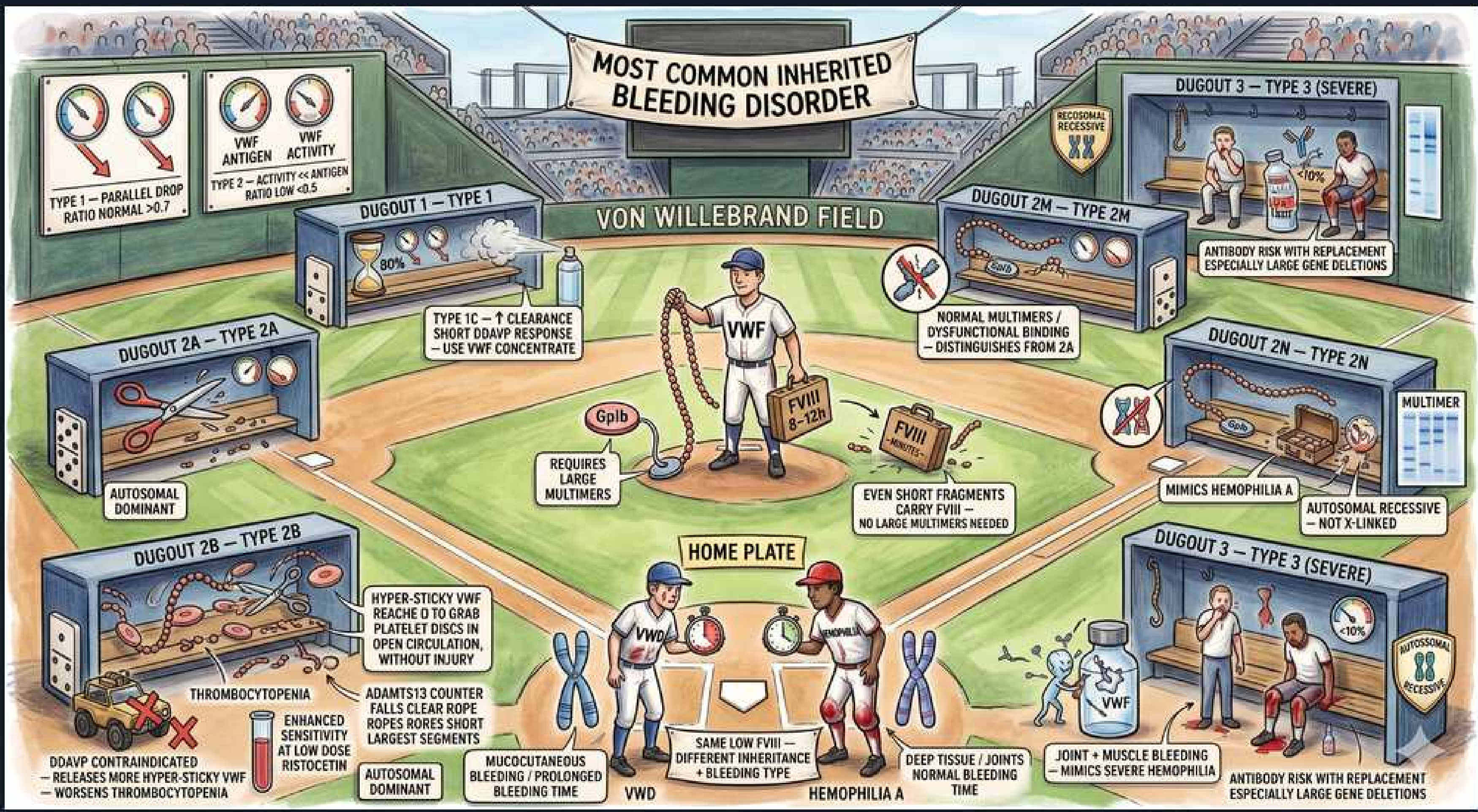
Platelet disorders — von Willebrand disease (Von Willebrand Field)



1. Most common inherited bleed

WHAT YOU SEE

The stadium banner 'MOST COMMON INHERITED BLEEDING DISORDER' over VON WILLEBRAND FIELD — the headline marquee of the whole scene.

WHAT IT ENCODES

vWD is the MOST COMMON inherited bleeding disorder (prevalence ~1% by lab criteria, ~0.1% symptomatic). Inheritance is usually autosomal DOMINANT (types 1 and most 2) — distinguishing it from X-linked hemophilia A/B. vWF is a large multimeric glycoprotein with two jobs: (1) it tethers platelets to subendothelial collagen via platelet GpIb (primary hemostasis) and (2) it carries/stabilizes Factor VIII in circulation. Loss of either function explains the mucocutaneous + secondary hemostatic bleeding pattern.

BOARD TRAP

WRONG: choosing hemophilia A as the most common inherited bleeding disorder. Hemophilia A is the most common X-linked/severe factor deficiency, but vWD is far more prevalent overall. Discriminator: vWD gives MUCOCUTANEOUS bleeding with autosomal inheritance (affects both sexes); hemophilia gives deep joint/muscle bleeds in males.

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2. Type 1 — partial quantity

WHAT YOU SEE

DUGOUT 1 — TYPE 1 showing the vWF activity and antigen gauges dropping in PARALLEL with 'RATIO NORMAL >0.7' — both fall together, so quality is intact.

WHAT IT ENCODES

Type 1 is the most common subtype (~75%): a PARTIAL QUANTITATIVE deficiency of structurally normal vWF. Both vWF antigen and vWF activity (ristocetin cofactor) fall in PARALLEL, so the activity/antigen ratio stays NORMAL (>0.7). Autosomal dominant with variable penetrance. Because the vWF that is present works normally, bleeding is usually mild-to-moderate and DDAVP works well (releases stored normal vWF from Weibel-Palade bodies).

BOARD TRAP

WRONG: assuming a low activity-to-antigen ratio in a mild bleeder means type 1. A ratio <0.7 signals a QUALITATIVE (type 2) defect, not the proportional type 1 drop. Discriminator: in type 1 the ratio is preserved; in type 2 activity is disproportionately low relative to antigen.

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3. vWF Ag vs activity ratio

WHAT YOU SEE

Two side-by-side dials, vWF ANTIGEN and vWF ACTIVITY, with 'RATIO <0.7' flagging type 2 — the activity dial reads disproportionately low versus antigen.

WHAT IT ENCODES

The vWF activity/antigen ratio is the key subtyping number. RATIO ≥0.7 = quantitative problem (type 1 or 3, normal-quality vWF). RATIO <0.7 = qualitative problem (type 2 — there is protein present but it does not function). Activity is measured as ristocetin cofactor activity (vWF:RCo) or the newer vWF:GpIbM assay; antigen is vWF:Ag by immunoassay. Always order vWF:Ag, vWF:RCo (activity), and FVIII together as the initial panel.

BOARD TRAP

WRONG: ordering only vWF antigen and calling it normal. A normal antigen with low activity (ratio <0.7) is exactly the type 2 pattern that an antigen-only test misses. Discriminator: you must measure ACTIVITY (ristocetin cofactor), not just antigen, to catch qualitative defects.

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4. vWF carries FVIII

WHAT YOU SEE

The central pitcher on the mound cradling a glowing FVIII coin labeled vWF — vWF is the carrier hauling and protecting Factor VIII.

WHAT IT ENCODES

vWF is the circulating CARRIER and stabilizer of Factor VIII — bound FVIII has a half-life of ~12 h, but unprotected FVIII is degraded within ~2 h. Therefore low or dysfunctional vWF causes a SECONDARY drop in FVIII, which can prolong the aPTT. PT is normal (extrinsic/common pathway intact). This is why severe vWD can mimic hemophilia A on coagulation testing despite a primary platelet-adhesion problem.

BOARD TRAP

WRONG: interpreting a prolonged aPTT with low FVIII as proof of hemophilia A. In vWD the low FVIII is SECONDARY to low vWF. Discriminator: check vWF:Ag/activity — they are low in vWD and normal in true hemophilia A; also vWD has prolonged PFA-100/bleeding time, hemophilia does not.

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5. Type 2A

WHAT YOU SEE

DUGOUT 2A — scissors cutting away the LARGE vWF MULTIMERS ('AUTOSOMAL DOMINANT'), losing the big hemostatic multimers.

WHAT IT ENCODES

Type 2A is the most common qualitative subtype: a DECREASE in high-molecular-weight (large) multimers, which are the most hemostatically active forms. Mechanism is either impaired multimer assembly/secretion or increased ADAMTS13 cleavage. Result: reduced platelet adhesion, low vWF:RCo with a LOW activity/antigen ratio (<0.7) but loss of large multimers on gel electrophoresis. Autosomal dominant. DDAVP response is variable.

BOARD TRAP

WRONG: treating absent large multimers as automatically type 2B. Both 2A and 2B lose large multimers, but only 2B has thrombocytopenia and an ENHANCED response to LOW-DOSE ristocetin (RIPA). Discriminator: low-dose ristocetin aggregation is INCREASED in 2B, normal/decreased in 2A.

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6. Type 2B gain-of-function

WHAT YOU SEE

DUGOUT 2B — HYPER-STICKY vWF grabbing platelets too eagerly ('REACH/GRAB PLATELET DISCS, THROMBOCYTOPENIA, ENHANCED LOW-DOSE RISTOCETIN, DDAVP CONTRAINDICATED') — the over-grabbing vWF that eats platelets.

WHAT IT ENCODES

Type 2B is a GAIN-OF-FUNCTION mutation: mutant vWF binds platelet GpIb spontaneously at high affinity, clearing both the platelet-vWF complexes and large multimers → mild THROMBOCYTOPENIA plus loss of large multimers. Hallmark lab: ENHANCED aggregation to LOW-DOSE ristocetin (RIPA). Bleeding worsens with stress, surgery, or pregnancy as more abnormal vWF is released.

BOARD TRAP

WRONG: giving DDAVP to a type 2B patient. DDAVP is CONTRAINDICATED in 2B — it releases more hyperadhesive vWF, worsening platelet clumping and thrombocytopenia. Discriminator: a vWD patient with low platelets + enhanced low-dose ristocetin response = 2B = use vWF concentrate, NOT desmopressin.

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7. Type 2N mimics hemophilia

WHAT YOU SEE

DUGOUT 2N — vWF that can't hold the FVIII coin ('MIMICS HEMOPHILIA A, AUTOSOMAL RECESSIVE') — the broken FVIII-binding site.

WHAT IT ENCODES

Type 2N ('Normandy') is a qualitative defect in the FVIII-BINDING site of vWF. vWF antigen and platelet function are NORMAL, but FVIII cannot be carried/stabilized → isolated LOW FVIII and prolonged aPTT. It is AUTOSOMAL RECESSIVE, so it affects both sexes and clinically mimics mild hemophilia A. Diagnosis: vWF:FVIII binding assay (low) plus genetic testing.

BOARD TRAP

WRONG: diagnosing hemophilia A in a FEMALE (or in a male without X-linked family pattern) who has isolated low FVIII. Discriminator: in 2N vWD the FVIII does not rise normally after FVIII infusion because there is no carrier; the vWF:FVIII-binding assay is abnormal — confirm before committing to a hemophilia diagnosis.

8. Type 3 severe

WHAT YOU SEE

DUGOUT 3 — TYPE 3 (SEVERE) with the vWF shelf essentially EMPTY/near-absent — the severe near-total deficiency.

WHAT IT ENCODES

Type 3 is the SEVERE form: virtually ABSENT vWF (antigen and activity near zero) with secondarily very LOW FVIII (often <10%). AUTOSOMAL RECESSIVE. Patients bleed like both severe vWD AND moderate-severe hemophilia — mucocutaneous bleeding PLUS hemarthroses and deep tissue bleeds. Treatment requires vWF-containing concentrate (DDAVP is useless — there are no stores to release).

BOARD TRAP

WRONG: choosing desmopressin (DDAVP) for type 3 vWD. With essentially no vWF stored in endothelium, there is nothing to release. Discriminator: type 3 → vWF/FVIII concentrate replacement; DDAVP only works when functional vWF stores exist (type 1).

9. Type 3 antibody risk

WHAT YOU SEE

The Type 3 SEVERE corner captioned 'ANTIBODY RISK WITH REPLACEMENT, ESPECIALLY LARGE GENE DELETIONS' — alloantibodies forming against infused vWF.

WHAT IT ENCODES

Because type 3 patients (especially those with large gene deletions/null alleles) have never been exposed to native vWF, repeated vWF replacement can trigger ALLOANTIBODIES (anti-vWF inhibitors). These antibodies neutralize concentrate AND can cause anaphylactic/anaphylactoid reactions on re-exposure. Management may then require FVIII without vWF or recombinant FVIIa for bleeds.

BOARD TRAP

WRONG: escalating the dose of standard vWF concentrate when a type 3 patient stops responding. The failure may be an alloantibody/inhibitor, not underdosing. Discriminator: poor recovery after replacement + history of gene deletion should prompt an inhibitor workup, not blind dose escalation.

10. Mucocutaneous bleeding

WHAT YOU SEE

HOME PLATE with the vWD chromosome player labeled 'MUCOCUTANEOUS BLEEDING, LONG BLEEDING TIME' — the nosebleed/bruise/menorrhagia surface-bleeding pattern.

WHAT IT ENCODES

vWD bleeding is classically MUCOCUTANEOUS, reflecting the primary-hemostasis (platelet adhesion) defect: epistaxis, easy bruising, gum/oral bleeding, MENORRHAGIA (often the presenting complaint in young women), and bleeding after dental work or surgery. Labs: prolonged PFA-100 closure time (modern replacement for bleeding time), normal platelet COUNT, normal PT, and aPTT normal-or-prolonged depending on FVIII level.

BOARD TRAP

WRONG: attributing a young woman's heavy menses + easy bruising to a platelet-count problem or just 'normal heavy periods.' A normal platelet count with prolonged PFA-100 and mucocutaneous bleeding points to vWD or a platelet-FUNCTION defect. Discriminator: deep joint/muscle bleeding instead suggests hemophilia, not typical vWD.

11. Hemophilia A contrast

WHAT YOU SEE

The HEMOPHILIA A chromosome player at home plate labeled 'DEEP TISSUE / JOINTS BLEEDING, SAME LOW FVIII' — the contrasting deep hemarthrosis, X-linked picture.

WHAT IT ENCODES

Contrast classic: HEMOPHILIA A is X-LINKED RECESSIVE (males affected), an ISOLATED Factor VIII deficiency, with DEEP bleeding — hemarthroses (target joints), muscle hematomas, and delayed post-surgical bleeding. Labs: prolonged aPTT, normal PT, normal platelet count, normal vWF, and normal PFA/bleeding time (primary hemostasis intact). Treatment is FVIII replacement (and DDAVP for mild disease, which raises FVIII).

BOARD TRAP

WRONG: expecting a prolonged bleeding time/PFA-100 in hemophilia A. Hemophilia spares primary hemostasis, so PFA-100 is NORMAL. Discriminator: prolonged PFA-100 + low vWF = vWD; normal PFA-100 + isolated low FVIII + X-linked male = hemophilia A.

12. Bernard-Soulier mimic

WHAT YOU SEE

The center note 'EVEN SHORT FRAGMENTS CARRY FVIII / NO LARGE MULTIMERS NEEDED' linking vWF to the platelet GpIb receptor — the receptor angle that makes Bernard-Soulier mimic vWD on ristocetin testing.

WHAT IT ENCODES

vWF works through platelet receptor GpIb, so receptor disorders mimic vWD on the ristocetin test. BERNARD-SOULIER syndrome (GpIb deficiency) shows ABSENT ristocetin-induced aggregation that, unlike vWD, does NOT correct with added normal plasma/vWF — plus giant platelets and thrombocytopenia. Glanzmann thrombasthenia (GpIIb/IIIa defect) instead has absent aggregation to ALL agonists EXCEPT ristocetin. These distinctions are high-yield on platelet-function questions.

BOARD TRAP

WRONG: calling absent ristocetin aggregation 'vWD' without a mixing/correction step. In vWD, adding normal plasma (supplying vWF) CORRECTS the ristocetin aggregation; in Bernard-Soulier it does NOT (the platelet receptor itself is defective). Discriminator: the normal-plasma mixing study separates vWD from a GpIb receptor disorder.